Swami Vivekānanda Yoga Anusandhāna Saṁsthāna (S-VYASA)

(Deemed to be University)

USN										

Examination	M1 Contest	**Academic Year**	2026–2027
Programme	B.Tech	**Specialization**	AI
Semester	I	**Date of Examination**	18/09/2026
Course Code		**Session**	_____
Course Name	Mathematics for Problem Solving		

Name: _____ **Programme/Section:** _____

Maximum Duration: 90 minutes **Maximum Marks:** 35

Instructions:

1. All questions are compulsory. Assume all variables are real unless stated otherwise.
2. In Section A, select one correct option; in Section B, write the required value or expression.
3. In Section C, match each statement with the correct numbered expression.
4. Show clear reasoning and working in Section D.

Section A: Multiple-Choice Questions

5 Q × 1 M = 5 M

Que. No.	Questions	CO	RBTL	Marks
1	A company sells a video game. If its price is p rupees, the company estimates that it will sell $20{,}000 - 500p$ games. Which expression represents the revenue? **A)** $(20{,}000 - 500p) + p$ **B)** $(20{,}000 - 500p) - p$ **C)** $\dfrac{20{,}000 - 500p}{p}$ **D)** $p(20{,}000 - 500p)$	CO1	2	**1**
2	The lines $y = x$ and $y = 3x$ pass through the origin. A student claims that the slope of a line bisecting the angle between them is $(1 + 3)/2 = 2$. Why is this reasoning incorrect? **A)** Slopes should be multiplied rather than averaged. **B)** An angle bisector averages the angles made with the x-axis, not their slopes; slope is the tangent of the angle. **C)** The correct slope must be $\sqrt{3}$, the geometric mean of the slopes. **D)** A line through the origin cannot have the form $y = mx$.	CO1	3	**1**
3	Let $p(x) = x^2 + ax + b$ have roots c and d. Define $$q(x) = x^2 + (2c + a)x + c^2 + ac + b.$$ Which of the following are roots of q(x)? **A)** Its roots are 0 and $d - c$. **B)** Its roots are c and $d + c$. **C)** Its roots are c^2 and d^2. **D)** It has the same roots as $p(x)$.	CO2	3	**1**

Que. No.	Questions	CO	RBTL	Marks
4	If $a < b$, which statement is always true? **A)** $-2a < -2b$ **B)** $a + 7 < b + 7$ **C)** $a^2 < b^2$ **D)** $\frac{1}{a} > \frac{1}{b}$	CO3	2	**1**
5	The graph of $y = x^2 - 4x + k$ touches the x-axis at exactly one point. Which condition must hold? **A)** The discriminant is negative. **B)** The discriminant is zero. **C)** $k = 0$. **D)** The roots have opposite signs.	CO2	2	**1**

Section B: Fill in the Blanks

5 Q × 1 M = 5 M

Que. No.	Questions	CO	RBTL	Marks
6	A falling object travels $d = 16t^2$ feet after t seconds. After 6 seconds, it has travelled _____ feet.	CO2	2	**1**
7	Aarav and Meera have positive integer ages x and y, and both are at least 3 years old. The clues $x + y = 12$ and "two years ago, the sum of their ages was 8" allow _____ different ordered pairs (x, y).	CO1	3	**1**
8	A square has side length x. Four 1×1 corner squares are removed. The area of the remaining region is _____.	CO1	2	**1**
9	If $\frac{1}{x} > 0$, then x is _____.	CO3	2	**1**
10	The equation $x^2 + (k-3)x + k = 0$ has equal roots. The sum of all possible values of k is _____.	CO2	3	**1**

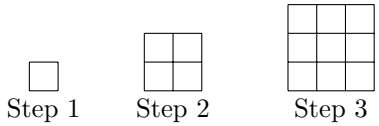
Section C: Match the Following

1 Q × 5 M = 5 M

Que. No.	Questions	CO	RBTL	Marks
11	A student club sends a fundraising appeal to n students and sets a target of T rupees. Match each description in Column A with its expression in Column B. **Column A** — **Column B** **A.** Amount raised on reaching half the target — **1.** $T + 500$ **B.** Amount raised if every student gives Rs. 50 — **2.** $\frac{T}{2}$ **C.** Amount raised if the club exceeds its target by Rs. 500 — **3.** $25n$ **D.** Amount still required after every student gives Rs. 50 — **4.** $50n$ **E.** Amount raised if half the students give Rs. 50 — **5.** $T - 50n$	CO1	2	**5**

Section D: Attempt the Following Questions

5 Q × 4 M = 20 M

Que. No.	Questions	CO	RBTL	Marks
12	Saira has the following pieces: • one square of side x units; • eight rectangular strips, each measuring $x \times 1$ units; and • fifteen squares, each of side 1 unit. She claims that all the pieces can be arranged, without gaps or overlaps, to form one large rectangle. **(i)** Write an algebraic expression for the total area of all the pieces. **(ii)** Use the expression to determine the length and breadth of the resulting rectangle in terms of x.	CO1	3	**4** **(2)** **(2)**
13	The first three steps of a growing square pattern contain 1, 4 and 9 small squares respectively.  Step 1 Step 2 Step 3 **(i)** Write an equation relating the step number n and the number of squares y, and explain how it represents the pattern. **(ii)** Determine the numbers of small squares in Steps 4 and 15.	CO1	3	**4** **(2)** **(2)**

<table>
<tr><th>Que. No.</th><th>Questions</th><th>CO</th><th>RBTL</th><th>Marks</th></tr>
<tr><td>14</td><td>Intelligence Quotient is calculated using
$$\text{IQ} = \frac{\text{Mental Age}}{\text{Chronological Age}} \times 100.$$
 

(i) For children of chronological age 15 years whose IQ satisfies $90 \leq \text{IQ} \leq 150$, determine the possible range of mental age.
(ii) For children of chronological age 12 years whose mental age satisfies $9 \leq \text{MA} \leq 15$, determine the possible IQ range.</td><td>CO3</td><td>3</td><td>4

(2)
(2)</td></tr>
<tr><td>15</td><td>A model rocket is launched from a platform. Its height above the ground after t seconds is
$$h(t) = -5t^2 + 20t + 25.$$
(i) Find the zeros of $h(t)$ and interpret the physically meaningful zero.
(ii) Find the maximum height without using calculus.
(iii) A student calculates $h(6) = -35$ and claims the rocket is 35 m underground. Is this interpretation correct? Explain.</td><td>CO2</td><td>3</td><td>4

(2)
(1)
(1)</td></tr>
<tr><td>16</td><td>At the Word Store, consonants are free, while each vowel A, E, I, O, U has its own fixed price. The following word prices are known:
<table><tr><td>Word</td><td>BANANA</td><td>MATRIX</td><td>CUBIC</td><td>PENCIL</td><td>ROBOT</td></tr><tr><td>Cost (Rs.)</td><td>9</td><td>7</td><td>9</td><td>10</td><td>4</td></tr></table>
Determine the cost of the word EDUCATION. Your solution must explain how the given word prices determine the required vowel prices; simply stating the final cost is not sufficient.</td><td>CO1</td><td>4</td><td>4</td></tr>
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